Bihar Engineering University, Patna

B.Tech 1st Semester Examination, 2025

Course: B.Tech | *Time: 03 Hours*
Code: 100111 | *Subject: Programming for Problem Solving* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Which software converts source code into object code?
(i) Interpreter (iii) Editor
(ii) Compiler (iv) Loader

(b) An algorithm is a
(i) Flowchart (iii) Step-by-step procedure to solve a problem
(ii) Programming language (iv) Hardware device

(c) Which operator is used for logical AND?
(i) & (iii) |
(ii) && (iv) ||

(d) Which operator has the highest precedence?
(i) + (iii) =
(ii) * (iv) %

(e) The index of the first element of an array is
(i) 0 (iii) -1
(ii) 1 (iv) Any number

(f) Which function is used to find string length?
(i) strcpy() (iii) strcat()
(ii) strlen() (iv) strcmp()

(g) A function calling itself is called
(i) Loop (iii) Iteration
(ii) Recursion (iv) Nesting

(h) Which type of parameter passing copies the value?
(i) Call by value (iii) Pointer passing
(ii) Call by reference (iv) Structure passing

(i) Which keyword is used to create a structure?
(i) struct (iii) typedef
(ii) union (iv) enum

(j) Which function is used to open a file?
(i) fopen() (iii) fclose()
(ii) fread() (iv) fprintf()

Q.2 (a) Explain the major components of a computer system such as disks, memory, processor, operating system, and compiler. Also describe where a program is stored and where it is executed. **[7]**

(b) Define algorithm and explain it's characteristics with an example? **[7]**

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Q.3	(a)	Define a variable. Explain different data types and describe how variables are stored in memory locations with suitable examples.	[7]
	(b)	Explain Syntax Errors and Logical Errors. Differentiate between them with suitable programming examples.	[7]
Q.4	(a)	Explain arithmetic operator and Assignment operator with a program.	[7]
	(b)	Explain Relational Operator with a program?	[7]
Q.5	(a)	Explain operator precedence and associatively with an example expression evaluation? X = 9 – 12 / 3 + 3 * 2 - 1	[7]
	(b)	Write the following C programs: (i) Write a C program to find greatest of three numbers by using ladder if else statement? (ii) Write a C program to find the total, average of given three integers?	[7]
Q.6	(a)	Explain array declaration and initialization in C. Also describe one-dimensional (1-D) arrays with suitable examples and programs.	[7]
	(b)	Explain two-dimensional (2-D) arrays in C. Describe their declaration, initialization, memory representation, and give a suitable example program	[7]
Q.7	(a)	Define function, and explain Function declaration, Function definition and Function call with example?	[7]
	(b)	Define recursion and explain with suitable example?	[7]
Q.8	(a)	Explain the following parameter passing techniques (i) Call-by-value (ii) Call-by-Reference	[7]
	(b)	Write a C program using a structure to store and display the details of a student such as Roll Number, Name, and Marks. The program should accept values from the user and display them properly.	[7]
Q.9		Write short note on ***any two*** of the following: (a) Switch statement (b) While loop (c) Pointer (d) Text file	**[7 x 2 = 14]**

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